

Euclid Data in the Cloud: Access, Analysis, and Science Opportunities

ENSCI (Euclid NASA Science Center at IPAC)

IRSA (NASA Infrared Science Archive)

248th AAS — June 14th, 2026



Today's Agenda

≈ 90 minutes — intro slides, then three notebooks in Fornax

1

Intro & framing

Workshop setup, Euclid + ENSCI, Q1 release, data-product acronyms, Fornax sign-up.

SLIDES

~ 15 min

2

Euclid data in the cloud

TAP / SIA / direct-S3 on a single EDF-N source — MER catalog, mosaic cutout, SIR 1D spectrum, SPE line features.

NOTEBOOK

~ 20 min

3

Galaxy clusters

Same toolkit on a high-richness Q1 cluster — ADQL JOIN across MER + PHZ, DBSCAN vs control field, over-density on H-band.

NOTEBOOK

~ 10 min

4

AGN hands-on

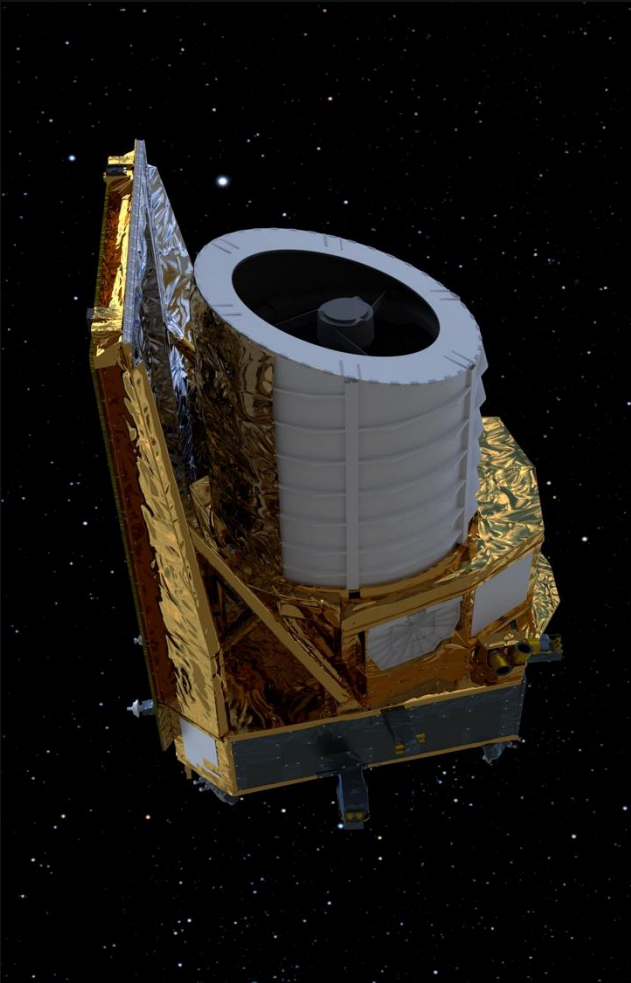
Project ~5k EDF-N AGN candidates onto a pre-trained color-SOM — compare selection methods, verify with three live cloud reads.

NOTEBOOK

~ 45 min

Euclid & ENSCI

ESA mission mapping the dark Universe — with NASA's US science center at IPAC



THE MISSION

Euclid

ESA M-class mission

Studies the geometry and nature of the dark Universe via weak gravitational lensing and galaxy clustering.

- **Launched** July 1, 2023 / 6-year mission at L2
- **1.2 m telescope** at optical + near-IR (0.5–2 μm)
- **14,000 sq deg** survey, a third of the sky
- **Consortium** of 2,500+ across 15 countries (Europe, US, Canada, Japan)

NASA

- **NASA contribution** 16 near-IR detectors + 3 nominated US science teams

THE US SCIENCE CENTER

ENSCI

Euclid NASA Science Center at IPAC

Established by NASA at Caltech/IPAC in Pasadena to support US-based investigators using Euclid data.

- **Operates SDC-US** one of 11 data processing nodes
- **US data archive** via NASA/IPAC IRSA
- **NISP data and Pipeline** via providing algorithm and software development, and data quality assurance
- **User support** Help Center, documentation, tutorials

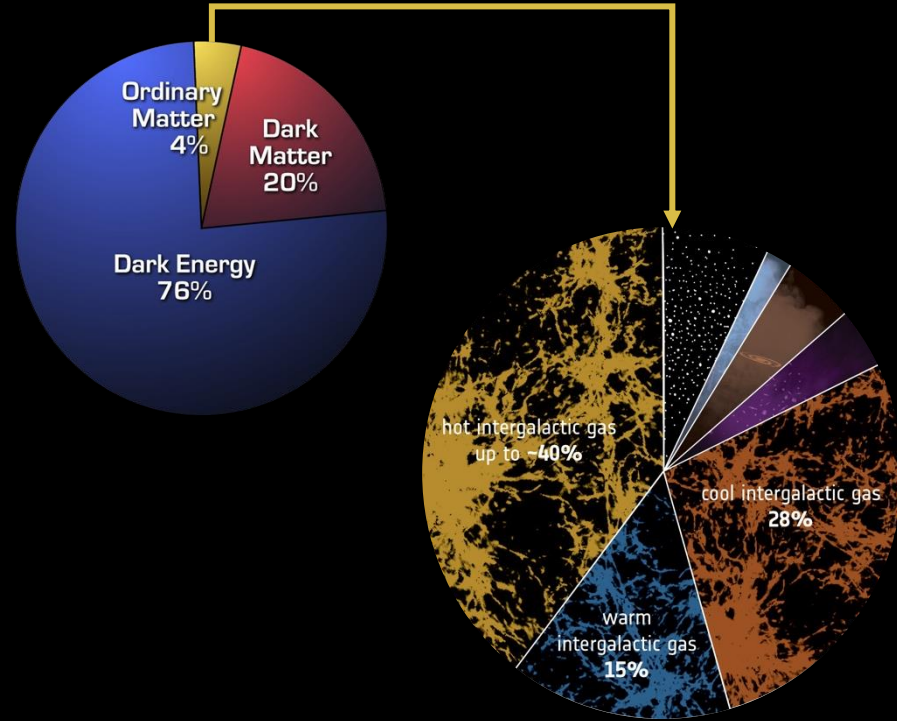
More at euclid.caltech.edu

Why Euclid? Mapping the Dark Universe

96% of the universe is dark — Euclid uses two complementary probes to measure it.

THE PROBLEM

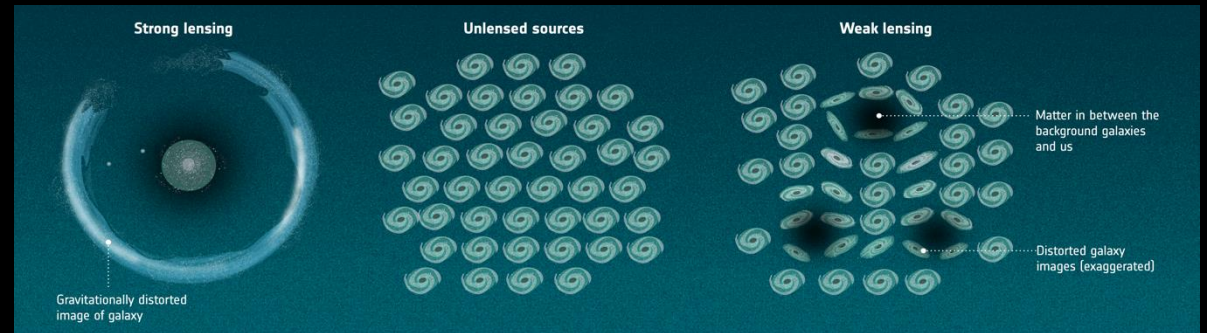
We don't know what 96% of the universe is



Only ~4% is ordinary matter. The rest, dark matter and dark energy.

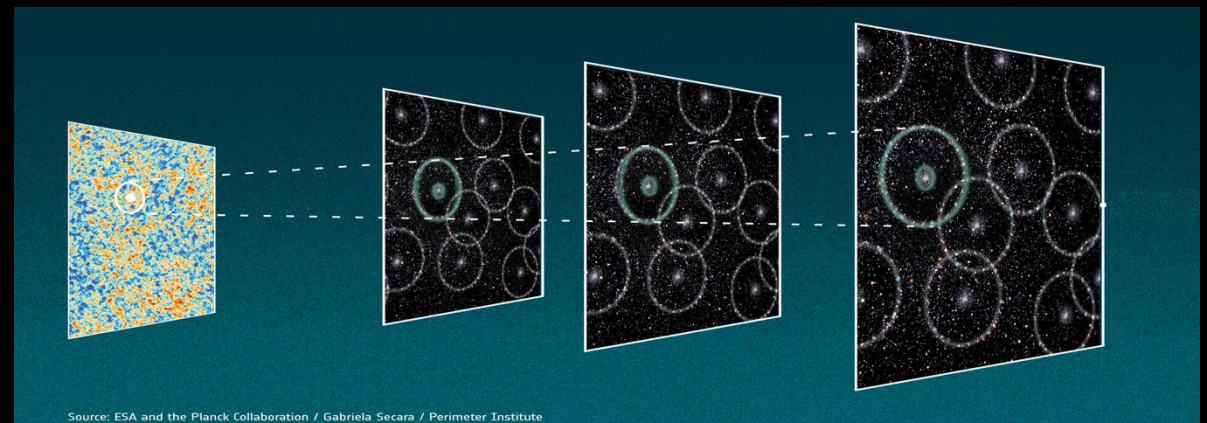
EUCLID'S TWO MAIN PROBES

1 Weak Lensing



Dark matter bends light from background galaxies, subtly distorting their shapes.

2 Galaxy Clustering (BAO/RSD)



Sound waves from the early universe imprint a "standard ruler" in galaxy clustering.

NISP — Photometric & Grism Redshifts

NIR photometry for $\sim 1.5\text{B}$ galaxies plus grism spectroscopy for $\sim 30\text{M}$ galaxies at $0.9 < z < 1.8$.

NIR PHOTOMETRY + SPECTROSCOPY

YJH bands $24 \text{ mag}_{\text{AB}}$ at 5σ

Red grism $1.25 - 1.85 \mu\text{m}$

Blue grism $0.92 - 1.30 \mu\text{m}$

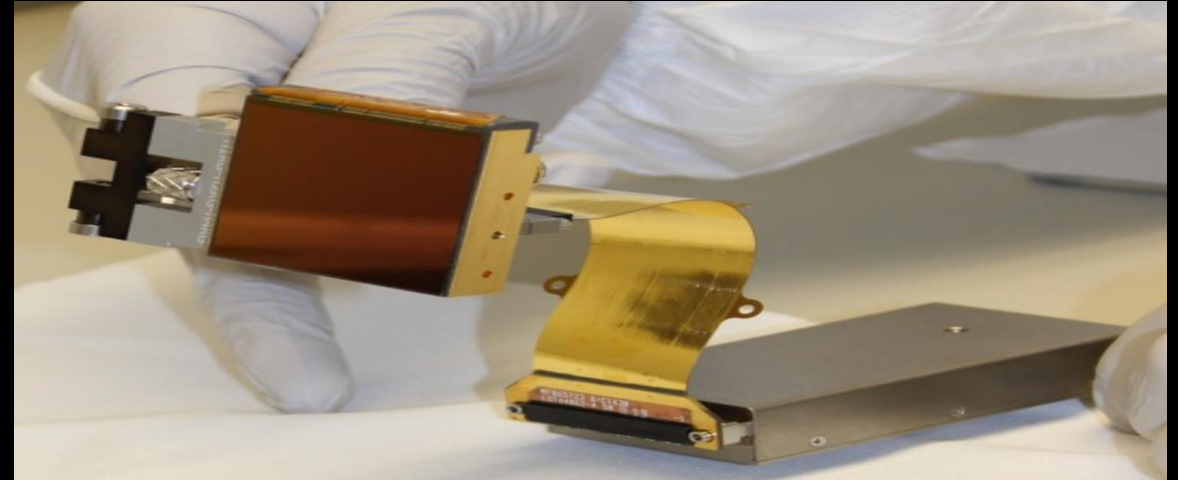
R(red grism) > 480 and R(blue grism) > 400

($0.5''$ source) · $0.3''$ / pixel

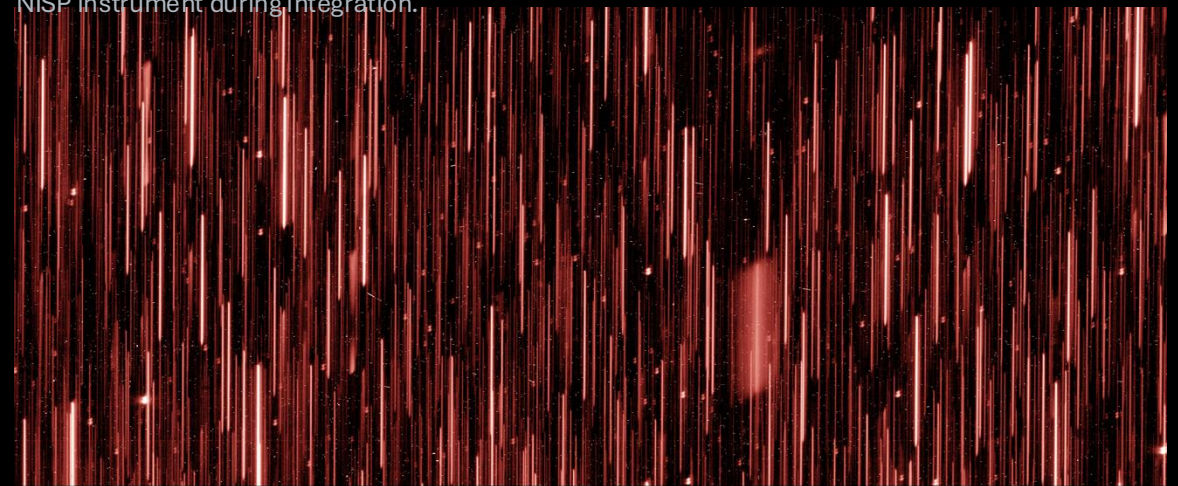
EARLY COMMISSIONING TEST IMAGE, NISP INSTRUMENT



Early commissioning test image (imaging mode).



NISP instrument during integration.



Early commissioning test image, NISP instrument (grism mode)

Commissioning test image (grism mode).

VIS — Shape Measurements

Wide-band optical imaging measures the shapes of ~1.5 billion galaxies for weak lensing.

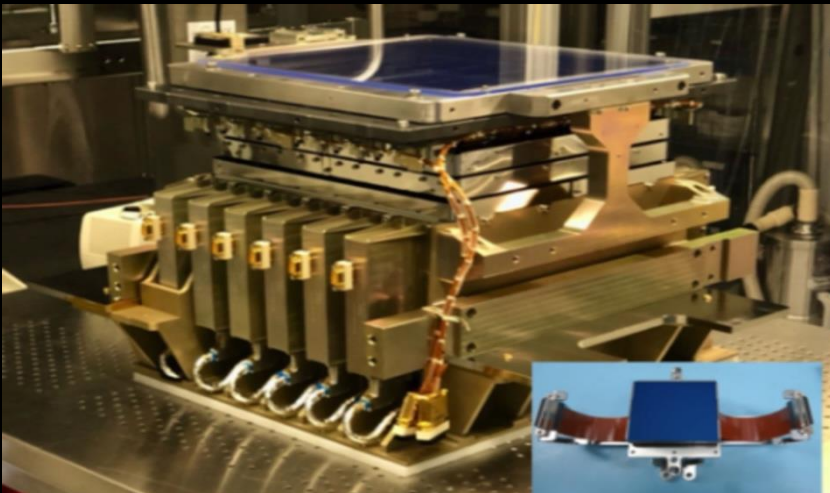
WIDE-BAND IMAGING

Bandpass 550 – 900 nm

Depth 24.5 mag_{AB} at 10 σ

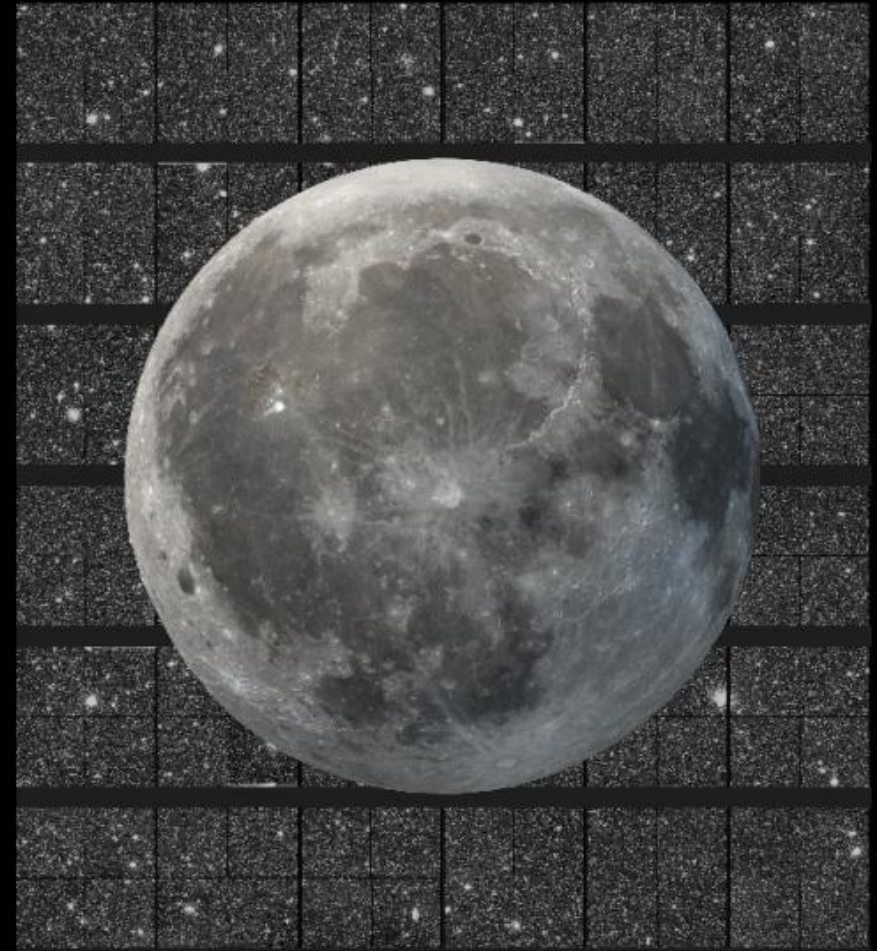
Pixel scale 0.1" / pixel

One broad VIS filter, optimized for high-resolution galaxy shape measurement.



VIS instrument during integration.

0.75 degree



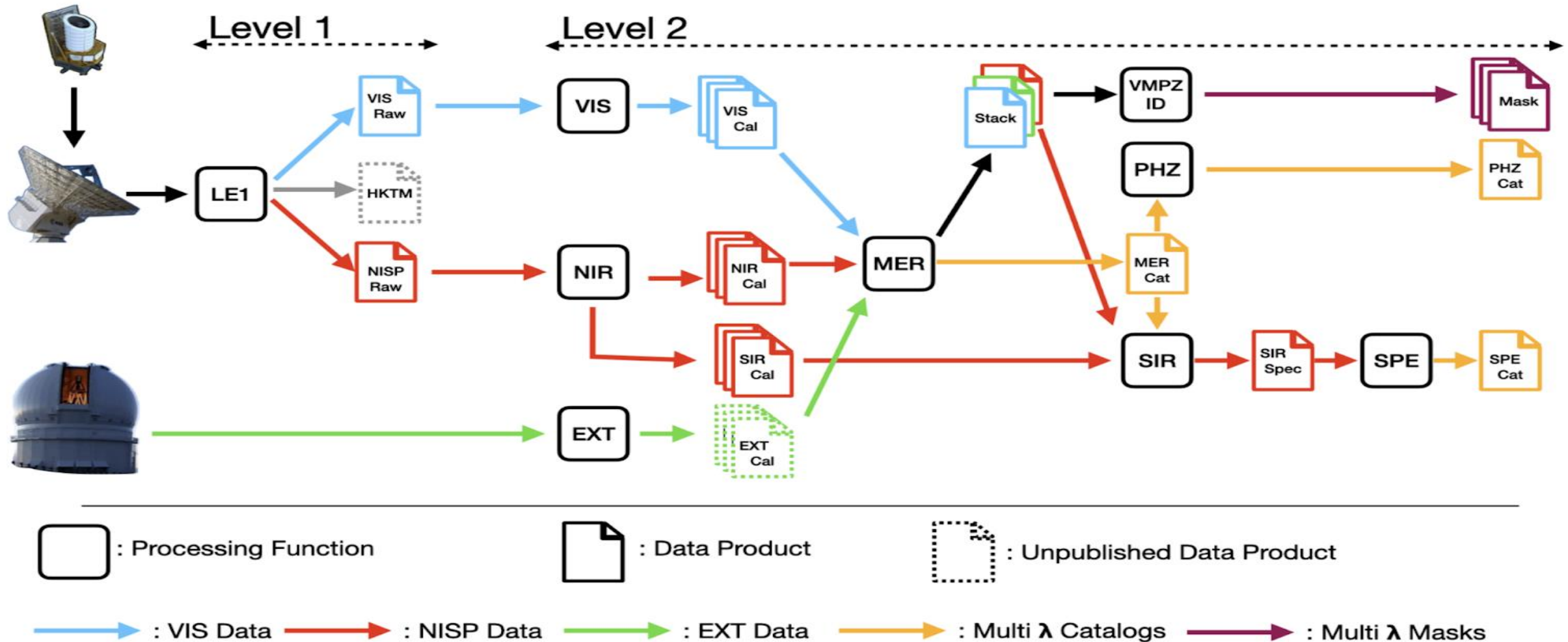
0.75 degree

VIS focal plane (36 CCDs) — the Moon shown for scale.

Data Products & Processing Flow

From Level 1 raw frames through Level 2 calibrated products to merged catalogs and masks.

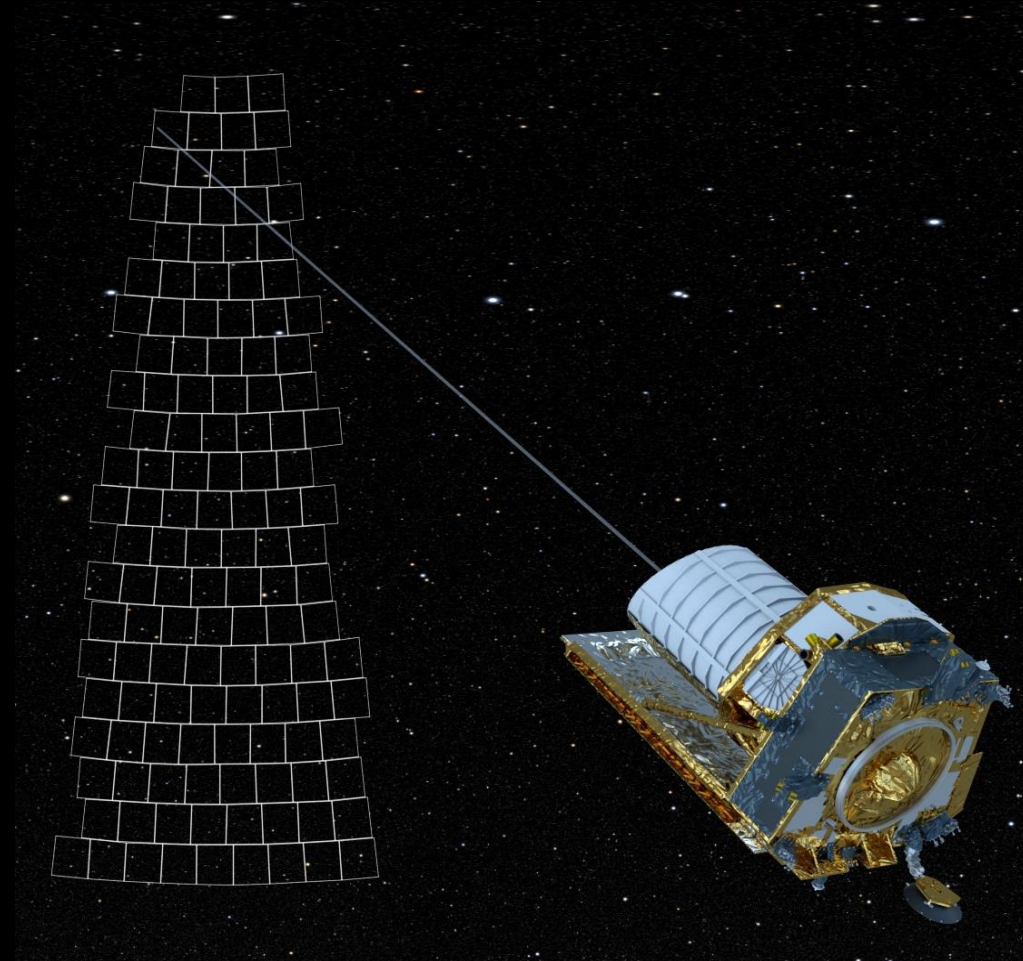
Euclid Collaboration: H. Aussel et al.: *Euclid* Quick Data Release (Q1): Overview



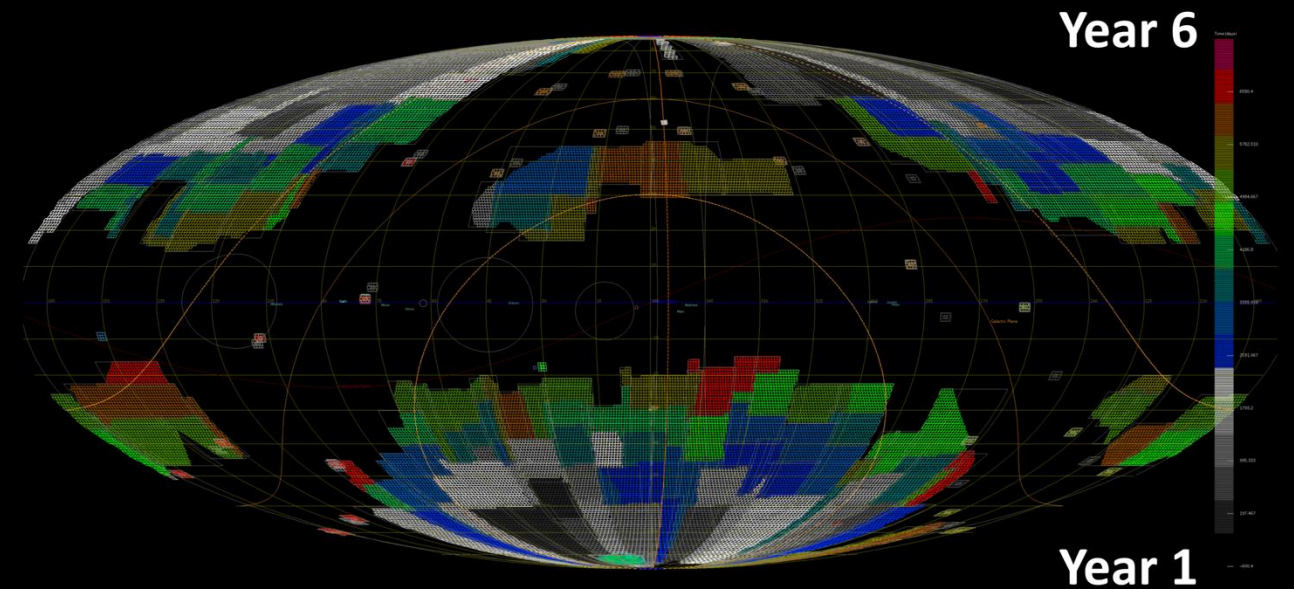
The Mission

The Euclid Survey

A $\sim 14,000 \text{ deg}^2$ wide survey plus three Euclid Deep Fields totaling $\sim 53 \text{ deg}^2$.



Step-and-stare observing — telescope tiles the sky one pointing at a time.



Wide Survey $\sim 14,000 \text{ deg}^2$ — color shows year of observation (Y1 $\rightarrow$ Y6).

THREE DEEP FIELDS

EDF-N

$20 \text{ deg}^2 \cdot \text{NEP}$

EDF-S

$23 \text{ deg}^2 \cdot \text{near SEP}$

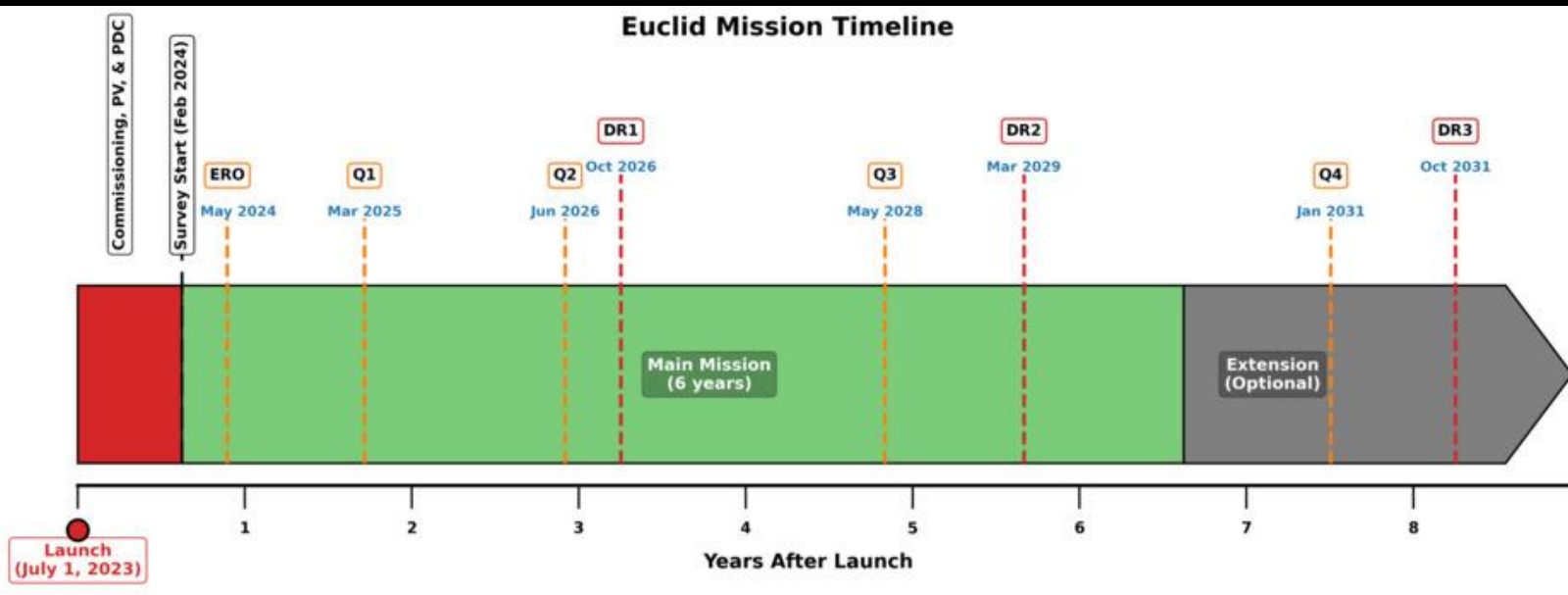
EDF-F

$10 \text{ deg}^2 \cdot \text{Fornax}$

+ *monthly 4 deg^2 Self-Calibration Field*

Data Release Schedule

Q1 released March 2025 · DR1 next



DR1 — DATA RELEASE 1

To be released: Fall 2026

~1900 deg²

depth ~ 24-24.5 mag (wide)

Access:

- NASA/IPAC IRSA
- ESA Euclid Science Archive

Q1 — QUICK RELEASE 1

Released March 19, 2025

~63 deg² over three Euclid Deep Fields

depth ~ 24-24.5 mag

- 23 deg² of **EDF South**
- 20 deg² of **EDF North**
- 10 deg² of **EDF Fornax**

Products:

- VIS + NISP imaging,
- NISP spectroscopy,
- catalogues & masks

Access:

- NASA/IPAC IRSA
- ESA Euclid Science Archive

Useful Links

ESA Q1 Data Release Page

cosmos.esa.int/web/euclid/euclid-q1-data-release

Euclid Q1 Data Product Description

euclid.esac.esa.int/dr/q1/dpdd

Euclid Tutorial Notebooks

caltech-ipac.github.io/irsa-tutorials/euclid

IRSA Euclid Webpage

irsa.ipac.caltech.edu/Missions/euclid.html

ENSCI Webpage

Euclid.Caltech.edu



NASA Fornax

JupyterLab in the cloud, next to NASA archive data

QUICK START

1. Open the quick-start guide

- docs.fornax.sciencecloud.nasa.gov/quick-start/
- or google “fornax quick start”

2. Make an account (approved instantly)

3. Log in to Fornax

4. Open a Jupyter Notebook

- Choose Medium • ~4 min to load

5. Click the “github extension”



6. Clone the workshop repo

- github.com/Caltech-IPAC/euclid-workshop
- Click “Download”



Scan to start

Quick-start guide

Q1 DATA RELEASE — MARCH 2025

63 deg² over three Euclid Deep Fields — VIS + NISP imaging, NISP spectroscopy, catalogues & masks.

FREE

BETA

BROWSER-BASED